

KSHITI TYAGI

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PROFESSIONAL SUMMARY

GenAI Engineer with 2+ years of production experience at Bosch building enterprise-grade AI solutions using RAG pipelines, LLMs, and AWS. Proven track record in reducing manual effort by 60% and improving model accuracy from 72% to 94%. Skilled in Agentic AI frameworks, LLM fine-tuning, MCP tool servers, and advanced prompt engineering. Currently building and deploying production AI systems targeting high-impact roles in GenAI and Data Analytics.

EXPERIENCE

Data Analyst & Software Engineer | Bosch Global Software Technologies | Oct 2023 – Oct 2025

- Designed and deployed an enterprise GenAI CoPilot chatbot using Python, RAG pipelines, and AWS Bedrock, enabling intelligent document processing across 8+ manufacturing plants and reducing manual processing effort by ~60%.
- Built RAG-based document intelligence pipelines using Python, TensorFlow, and LLM orchestration, improving data extraction accuracy from 72% to 94% and accelerating decision-making for plant managers and engineering teams.
- Processed unstructured PDFs and complex documents (text, images, tables) using the Unstructured library; delivered via a Streamlit app integrated with Azure Data Lake for centralized storage and real-time document processing.
- Built scalable LLM inference workflows with Amazon SageMaker enabling real-time analytics for 500+ engineers.
- Developed reusable Python automation scripts and standardized parsing frameworks across 8 manufacturing plants, reducing operational errors by 45% and manual intervention significantly.
- Partnered with global cross-functional stakeholders to refine chatbot workflows, incorporate user feedback, and ensure enterprise-level scalability and production-grade reliability.

Self-Directed Learning & Freelance | Independent AI/ML Projects & Upskilling | Nov 2025 – Present

- Developed an AI-Driven Agri-Intelligence Platform using geospatial ML for crop suitability and yield prediction with Amazon SageMaker and Milvus Vector Database.
- Built OCR-Based Aadhaar Verification System automating extraction of identity details from Aadhaar card images using Tesseract OCR, OpenCV preprocessing, regex-based structured field parsing, and FastAPI backend.
- Built Sales Intelligence Engine — end-to-end cloud-native ML pipeline for churn prediction and sales opportunity scoring using Snowflake, Python, and ETL workflows.
- Deepened expertise in Agentic AI, LLM fine-tuning, and multi-modal GenAI architectures.

Full-Stack Developer Intern | HighRadius | Jan 2022 – Apr 2022

- Built AI-powered B2B invoice automation platform using Python, Flask, and ML models to streamline billing workflows.
- Designed anomaly detection and classification models; reduced late payments by 30%.
- Created executive dashboards with Tableau to surface real-time business insights for stakeholders.

PROJECTS

GenAI CoPilot for Bosch Plants | Python · TensorFlow · RAG · AWS Bedrock · Azure Data Lake · Streamlit

- Designed and deployed an enterprise GenAI CoPilot chatbot to automate extraction of critical data from PDFs and structured documents, reducing manual processing effort by ~60% across multiple manufacturing plants.
- Built RAG-based document intelligence pipelines with LLM orchestration enabling 94% data extraction accuracy and context-aware information retrieval at scale.
- Integrated with Azure Data Lake and cloud-native data engineering pipelines for centralized storage, scalable ingestion, and real-time document processing.
- Delivered the system through a Streamlit app; collaborated with global teams to ensure enterprise-level scalability and seamless integration.

HR AI Hybrid Assistant | Python · LangChain · FAISS · HuggingFace · FastMCP · Streamlit

Live: hr-ai-hybrid-assistant.streamlit.app | GitHub: github.com/robo9900/hr-ai-hybrid-assistant

- Designed and deployed a production-grade HR AI assistant using a hybrid routing architecture — intelligently routing queries across RAG retrieval, MCP tool execution, and hybrid mode in real time.
- Built RAG pipeline using FAISS vector store and HuggingFace embeddings for instant HR policy retrieval with semantic search accuracy.
- Implemented MCP tool server (FastMCP) for transactional actions including leave application, approval, and balance tracking — enabling the system to execute actions, not just answer questions.
- Delivered ChatGPT-style UI with dark/light mode, employee login, leave balance tracker, and HR query log with CSV export; deployed on Streamlit Cloud with GitHub CI.

OCR-Based Aadhaar Verification System | *Python · FastAPI · OpenCV · Tesseract OCR · Regex*

- Built an OCR-based Aadhaar verification system to automate extraction of key identity details from Aadhaar card images using Computer Vision and FastAPI.
- Implemented image preprocessing using OpenCV and text extraction using Tesseract OCR to improve accuracy on uploaded document images.
- Developed regex-based parsing logic to extract structured fields (Aadhaar number, DOB, gender, PIN code) with Aadhaar masking for privacy; integrated Swagger API testing interface for real-world verification workflows.

AI-Driven Agri-Intelligence Platform | *Python · Amazon SageMaker · Milvus · Geospatial ML*

- Designed a geospatial ML solution for crop suitability and yield prediction across diverse landscapes.
- Built geospatial pipelines and trained models using Amazon SageMaker and Python; integrated Milvus Vector Database for spatial feature retrieval and field-level recommendations.

Sales Intelligence Engine | *Python · Snowflake · ETL · ML*

- Built a unified predictive analytics platform for churn prediction and sales opportunity scoring for commercial teams.
- Implemented ETL workflows, feature engineering, and model orchestration using Snowflake and Python; enabled model explainability, monitoring, and automated scoring across pipelines.

SKILLS

AI/ML: LLMs, GPT-4, Llama 3, RAG Architecture, Agentic AI, MCP Tool Servers, NLP, Deep Learning (TensorFlow, PyTorch), Prompt Engineering, LLM Fine-Tuning

Programming: Python, SQL

Cloud & MLOps: AWS Bedrock, Amazon SageMaker, Azure ML Studio, GCP, Docker, Kubernetes

Data & Databases: Snowflake, FAISS, Milvus, Pinecone, Azure Data Lake, ETL Pipelines

APIs & Tools: FastAPI, FastMCP, LangChain, Streamlit, OpenCV, Tesseract OCR, Git, Power BI, Tableau

EDUCATION

B.Tech in Computer Science | SRM University | 2023 | GPA: 8.64 / 10

KEY ACHIEVEMENTS

- Improved data extraction accuracy from 72% to 94%, strengthening reporting reliability at Bosch.
- Reduced manual processing by 60% via standardized GenAI workflows and process automation.
- Standardized Python automation across 8 manufacturing plants, cutting operational errors by 45%.
- Deployed live GenAI product (HR AI Hybrid Assistant) — publicly accessible on Streamlit Cloud with full RAG + MCP architecture.